

TEEN BIRTH PROJECT DESIGN DOCUMENT

Adolescent Birth Rates, High-School Dropout, and Demographic/Socioeconomic Disparities

A State-by-Year-by-Demographic Data Science Study

Item	Proposed Plan
Project type	Observational, population-level data science / statistical study
Geographic unit	U.S. states + District of Columbia, subject to source availability
Primary population	Adolescents / young adults, with teen births defined using mothers ages 15–19
Primary exposure	Teen fertility rate / adolescent birth rate
Primary outcome	High-school status dropout rate among ages 16–24
Key stratifiers	Race/ethnicity, state, year/period
Additional factors	Socioeconomic indicators such as poverty and health insurance, subject to data availability
Core methods	Data integration, exploratory analysis, association analysis, regression, and interpretable machine learning
Important interpretation	The study estimates associations and predictive relationships; it does not establish causality

1. Executive Summary

This project will examine whether places and demographic groups with higher adolescent birth/fertility rates also show higher high-school dropout rates, and whether the observed relationship differs across race/ethnicity, geography, and socioeconomic conditions. The project combines publicly available U.S. data, with CDC WONDER Natality data as the primary source for teen fertility/birth measures and NCES/Census-based education data for dropout measures. Additional American Community Survey (ACS) indicators may be incorporated to represent contextual socioeconomic conditions.

The project is designed as an observational study. Results will be framed as patterns, associations, disparities, and prediction rather than proof that teen pregnancy causes students to leave school. A final modeling stage will compare interpretable statistical and machine-learning approaches and use feature-importance/SHAP analysis to identify which available factors contribute most to model predictions.

2. Overall Goal

To analyze the relationship between adolescent birth/fertility rates and high-school dropout rates across demographic and geographic groups, while examining socioeconomic factors that may help explain differences in educational outcomes.

3. Main Research Question

Are states and demographic groups with higher adolescent birth/fertility rates also characterized by higher high-school dropout rates, and how does this relationship vary across race/ethnicity, geography, time, and socioeconomic context?

4. Specific Research Questions

1. RQ1: What are the trends in adolescent fertility/birth rates across states and demographic groups over the study period?
2. RQ2: What are the trends and disparities in high-school status dropout rates across states and racial/ethnic groups?
3. RQ3: Is adolescent fertility/birth rate statistically associated with high-school dropout rate at the state-demographic level?
4. RQ4: Does the strength or direction of the association differ across racial/ethnic groups or geographic regions?
5. RQ5: Do socioeconomic variables such as poverty and health-insurance coverage help explain differences in dropout rates?
6. RQ6: Can dropout rates be predicted from adolescent fertility/birth rates and contextual socioeconomic variables?
7. RQ7: Which features contribute most to model predictions, and are prediction errors materially different across demographic groups?

5. Working Hypotheses

1. H1: Higher adolescent fertility/birth rates will be positively associated with higher high-school status dropout rates.
2. H2: The strength of the association will differ across racial/ethnic groups.
3. H3: Socioeconomic context will explain part of the observed relationship between adolescent fertility and dropout rates.
4. H4: Models using both adolescent fertility and socioeconomic variables will perform better than a model using adolescent fertility alone.
5. H5: Predictive performance and/or error patterns may differ across demographic groups; these differences will be measured rather than assumed.

6. Data Sources

The following sources are proposed. Exact years and variables must be finalized after checking source availability and comparability.

Source	Purpose	Key variables	Important notes
CDC/NCHS – CDC WONDER Natality 2016–2024 Expanded	Primary adolescent birth/fertility measure	Year; state of mother's residence; mother's age; mother's single race; Hispanic origin; births; female population; fertility rate	CDC WONDER supports age 15–19 and race stratification. Fertility rates are normalized to females ages 15–44 and can be subset by mother's age, race, Hispanic origin, location, and year.
NCES Digest / Condition of Education	High-school dropout outcome	Status dropout rate; state/jurisdiction; race/ethnicity; year/period	State-by-race tables exist for specific years and multi-year averages. Availability is not identical to CDC annual coverage.
U.S. Census Bureau – ACS	Socioeconomic/contextual covariates and/or supplemental dropout estimation	Poverty; insurance; education; demographic characteristics; population denominators	ACS 1-year and 5-year products have different geographic and precision characteristics. The exact product must be chosen to match the analytical design.

7. Source-Specific Data Plan

7.1 CDC WONDER Natality

The current CDC WONDER query has been successfully configured to return state-level fertility results by mother's single race. The project should use the expanded Natality dataset for years that are ultimately compatible with the education outcome dataset.

1. Group Results By: State of Residence, Year, Mother's Single Race 6.
2. Mother's age: 15–19 years.
3. Mother's Hispanic Origin: All Origins initially; a separate Hispanic-origin analysis may be added if justified.
4. Measure: Births and Fertility Rate per 1,000; retain Female Population where available.
5. Residence: all U.S. states/DC, subject to source coverage.
6. Do not request Birth Rate when filtering by mother's age; CDC documentation states that birth-rate denominators are not available under those additional age restrictions.
7. Record suppression/unknown categories and do not silently treat suppressed cells as zero.

CDC documentation states that the expanded 2016–2024 Natality data include maternal age, state/county residence, single-race categories, Hispanic origin, and fertility rates. It also states that fertility rates are calculated using births divided by the number of females ages 15–44, with compatible subsetting by maternal age, race, Hispanic origin, residence, and year.

7.2 Education / Dropout Data

The outcome should be the NCES definition of status dropout rate: the percentage of 16–24-year-olds who are not enrolled in school and have not completed a high-school program. The team should distinguish this measure from event dropout rate and graduation rate.

1. Primary outcome label: Status_Dropout_Rate.
2. Retain standard errors/margins of error whenever the source provides them.
3. Preserve NCES suppression and reliability flags such as estimates that do not meet reporting standards.
4. Do not mix annual estimates and five-year-average estimates in one model without explicitly modeling the difference.
5. A key design decision is required: either use an NCES state-by-race multi-year dataset as the primary comparable outcome, or compute compatible ACS-based estimates directly for a consistent annual panel.
6. NCES examples show state-by-race status dropout tables for 2019 and 2021 and a 2017–2021 five-year-average table. Therefore, the final time window should be selected after the team verifies the exact dropout data available for all required state/race cells.

7.3 ACS Socioeconomic Variables

1. Possible variables: poverty rate, health-insurance coverage, educational attainment, unemployment/labor-force context, and relevant population/demographic controls.
2. Use a consistent ACS product and geography across years.
3. Document the exact table/variable IDs and universe for every selected measure.
4. If a variable is not directly comparable across years, either harmonize it carefully or exclude it from the longitudinal model.
5. Avoid adding many correlated socioeconomic variables without a conceptual reason; use a small, defensible set.

8. Recommended Analytical Dataset

The preferred analytical structure is a long-format panel in which each row represents one state × year/period × race/ethnicity unit. The final structure depends on the chosen education data design.

Variable	Description	Type	Example
state	State name or standardized state code	Categorical	Maryland / 24
year	Calendar year or reference period	Integer/categorical	2019
race_ethnicity	Harmonized demographic category	Categorical	Black
teen_fertility_rate	Fertility rate for mothers ages 15–19	Numeric	28.77
births_teen	Number of births to mothers ages 15–19	Integer	12651

female_population_15_44	CDC denominator associated with fertility rate	Integer	439782
dropout_rate	High-school status dropout rate for ages 16–24	Numeric (%)	5.9
dropout_se	Standard error where available	Numeric	0.52
poverty_rate	Selected ACS poverty indicator	Numeric (%)	TBD
insurance_rate	Selected ACS health-insurance indicator	Numeric (%)	TBD
population	Relevant population control	Numeric	TBD

9. Data Harmonization Strategy

1. Use state FIPS codes as the primary geographic key rather than state names alone.
2. Create a single standardized race/ethnicity field only after documenting how each source defines race and Hispanic origin.
3. Do not assume CDC 'single race' categories are identical to NCES race/ethnicity categories. The harmonization mapping must be documented.
4. Keep an 'original_category' field so source categories can be recovered.
5. Use a 'data_source' and 'source_year/period' field for auditability.
6. Distinguish missing, suppressed, not applicable, and unreliable estimates.
7. Do not replace suppressed values with zero.
8. Check duplicates after every merge using state + year/period + demographic group.
9. Create a data dictionary before modeling.

10. Analysis Plan

10.1 Phase 1 – Data Quality and Descriptive Analysis

1. Count observations by state, year/period, and demographic group.
2. Measure missingness and suppression.
3. Check distributions and extreme values.
4. Verify that rate units are consistent: CDC fertility rate per 1,000 versus NCES dropout rate in percent.
5. Produce summary statistics overall and by race/ethnicity.

10.2 Phase 2 – Trend and Disparity Analysis

1. Plot adolescent fertility rates over time by state and demographic group.
2. Plot dropout rates over time/period were comparable.
3. Compare highest and lowest states, with uncertainty and suppression rules respected.

4. Compare racial/ethnic gaps using absolute and relative differences where appropriate.
5. Create state maps only when the geographic measure and period are directly comparable.

10.3 Phase 3 – Association Analysis

1. Pearson correlation as an initial descriptive measure when assumptions are reasonable.
2. Spearman correlation as a robustness check for monotonic relationships and non-normal data.
3. Bivariate regression: $\text{dropout_rate} \sim \text{teen_fertility_rate}$.
4. Multivariable regression: $\text{dropout_rate} \sim \text{teen_fertility_rate} + \text{socioeconomic variables} + \text{year/period}$.
5. If enough repeated observations exist, consider state fixed effects and year fixed effects.
6. Cluster standard errors by state if repeated state observations are used.
7. Run sensitivity analyses with and without high-leverage/small-sample observations.

10.4 Phase 4 – Machine Learning

1. Start with a simple baseline such as mean prediction and linear regression.
2. Compare interpretable linear/regularized models with tree-based models such as Random Forest and, if justified, XGBoost.
3. Use train/test splitting that prevents information leakage. For panel data, consider time-based or group-based splits rather than random row splitting.
4. Evaluate using MAE, RMSE, and R^2 .
5. Compare model performance across demographic groups when sample sizes permit.
6. Do not optimize for complexity if a simpler model provides similar performance and better interpretability.

10.5 Phase 5 – Explainability and Fairness/Disparity Checks

1. Use SHAP or another feature-attribution method to explain the final tree-based model.
2. Report global feature importance and selected group-specific explanations.
3. Compare error metrics across demographic groups when sufficient data exist.
4. Check whether a model systematically overpredicts or underpredicts for particular groups.
5. Frame these as predictive disparity checks, not as proof of discrimination or causal mechanisms.

11. Statistical and Modeling Caveats

1. Ecological inference: state/race/year aggregates describe groups and places, not individual students.
2. Temporal alignment: CDC provides annual natality data, while NCES state-by-race dropout data may be annual for some years and five-year averages for others.
3. Different populations: teen fertility concerns mothers ages 15–19, while the NCES status dropout measure concerns persons ages 16–24.
4. Race/ethnicity definitions differ by source and must be harmonized transparently.

5. Suppression and small sample sizes can remove or destabilize some state-by-race estimates.
6. Socioeconomic variables may be highly correlated with one another, creating multicollinearity.
7. Observed associations may reflect unmeasured factors such as school context, labor markets, family structure, health access, or policy.
8. The project should not claim that teen pregnancy causes dropout based on this design alone.

12. Work Division – Three Students

The team should divide the project by workflow stage while requiring all three students to participate in the literature review, methodology discussion, interpretation, and final paper. This avoids creating three disconnected mini-projects.

Student	Primary role	Responsibilities	Main deliverables
Student 1	Data Engineering & Data Quality	CDC WONDER extraction; NCES/ACS extraction support; source documentation; cleaning; race/state harmonization; missing/suppression handling; master dataset; data dictionary.	Raw-data folder; cleaned datasets; data dictionary; cleaning notebook; merge/audit report.
Student 2	Statistics, EDA & Visualization	Descriptive statistics; trend analysis; state/race comparisons; correlation; regression diagnostics; publication-quality figures and tables.	EDA notebook; statistical tables; figures; interpretation notes.
Student 3	Machine Learning & Explainability	Baseline and ML models; train/test design; model evaluation; feature importance; SHAP; group-wise error/fairness checks; model interpretation.	Modeling notebook; model comparison table; SHAP figures; error/disparity analysis.

13. Shared Responsibilities

1. All students: literature review and identification of related work.
2. All students: weekly progress updates and reproducible code practices.
3. All students: review the master dataset before analysis.
4. All students: contribute to methodology and limitations.
5. All students: review figures/tables for consistency.
6. All students: contribute to the final manuscript and presentation.
7. One student may be designated as repository maintainer, but no student should be the only person who understands the pipeline.

14. Project Repository Structure (might change as per the requirements)

```
project/
├── data/
│   ├── raw/
│   ├── interim/
│   └── processed/
├── notebooks/
│   ├── 01_data_collection_cleaning.ipynb
│   ├── 02_eda_statistics.ipynb
│   └── 03_modeling_explainability.ipynb
├── src/
│   ├── data_cleaning.py
│   ├── harmonization.py
│   ├── analysis.py
│   └── modeling.py
├── figures/
├── tables/
├── results/
├── docs/
│   ├── data_dictionary.xlsx
│   └── source_log.xlsx
├── README.md
└── requirements.txt
```

15. Reproducibility Requirements

1. Record source URL, dataset name, query settings, download date, and exact years/periods for every dataset.
2. Save the exact CDC WONDER query configuration or screenshots/exported criteria.
3. Never manually edit raw data files.
4. Perform transformations through code/notebooks.
5. Use a fixed random seed for machine-learning experiments.
6. Save package versions in requirements.txt or an environment file.
7. Keep a changelog when variables or inclusion criteria change.
8. Generate final tables and figures from the processed dataset rather than manual spreadsheet edits.

16. Proposed Timeline (Might change)

Week	Milestone	Student 1	Student 2	Student 3
Week 1	Finalize research design	Confirm sources/variables	Finalize RQs/statistics	Finalize modeling plan

Week 2	Data acquisition	Download/extract raw data	Review variable definitions	Review model-ready requirements
Week 3	Cleaning harmonization +	Build master dataset	QA checks	QA checks
Week 4	EDA	Data-quality report	Descriptive analysis/figures	Baseline model
Week 5	Association analysis	Support final dataset	Correlation/regression	Feature engineering
Week 6	ML + explainability	Pipeline support	Interpret statistical results	ML, SHAP, error analysis
Week 7	Integration	Finalize data/methods	Finalize tables/figures	Finalize model results
Week 8	Paper presentation +	Methods/data sections	Results/figures	ML/results/discussion

17. Expected Outputs

1. A documented, reproducible state $\times$ year/period $\times$ demographic analytical dataset.
2. A complete data dictionary and source log.
3. Descriptive statistics and disparity tables.
4. Trend, correlation, and state/demographic comparison figures.
5. Regression results with appropriate uncertainty and diagnostics.
6. Baseline and machine-learning model comparison.
7. SHAP/global feature-importance visualizations.
8. Group-wise predictive error/disparity analysis where sample sizes support it.
9. A final research paper and presentation.
10. A reproducible GitHub-style project repository.

18. Success Criteria

Area	Minimum success criterion
Data	All included observations have documented source, state, year/period, demographic category, and variable definitions.
Quality	Suppression, missingness, duplicates, and outliers are explicitly assessed.
Analysis	At least one descriptive, one association, and one multivariable analysis are completed.
Modeling	At least one interpretable baseline and one stronger nonlinear model are compared.
Explainability	Model predictions are explained with feature importance/SHAP where appropriate.
Disparity	Results are examined across demographic groups when data support reliable comparison.
Reproducibility	Another team member can rerun the analysis from the documented raw-data steps.

Interpretation	Conclusions clearly distinguish association/prediction from causation.
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19. Decisions to be made

Before full-scale data collection, the following decisions should be confirmed:

1. Final study period: Should the primary analysis use 2016–2021, 2017–2021, or another period based on the exact dropout data available?
2. Dropout data design: Use published NCES state-by-race estimates/5-year averages, or calculate compatible estimates from ACS PUMS?
3. Demographic scope: Which race/ethnicity categories are sufficiently comparable across CDC and NCES/ACS?
4. Socioeconomic variables: Which 2–5 contextual variables are theoretically justified and available consistently?
5. Primary outcome: State-level status dropout rate, or a more education-specific outcome if a better aligned dataset is identified?
6. Modeling objective: prediction of dropout rate, explanation of association, or both?
7. Fairness objective: group-wise predictive error analysis only, or a formal fairness framework?